

You also want to highlight the cultural norms and values that make GitLab unique since contingent workers will still need to be aware of and adapt to these norms.

Interview Process

For the interview, you will have a lot of agency to structure interviews as an engineering manager looking to make this hire. It is advisable to stay as close as possible to the company-level guidelines, especially as it relates to diversity inclusion and belonging. You can also use your own sub-department or group guidelines as a basis for defining the interview process, including the diversity of interviewers and approach to technical vetting.

Contracting and invoicing

Once you have identified the candidate that you want to hire (either directly or through an agency), the actual process of contracting is based on the professional services procurement process outlined by the finance department.

You can read more in the handbook section above, but roughly the steps involved are as follows:

- Select a start and end date for the contract (Note: staff augmentation workers have a max of 24 months)
- Obtain (or draft yourself) a Statement of Work (SOW) outlining the work that the contingent workers will perform for GitLab and the terms and conditions as related to this contract. The SOW should specify the dates, the name of the individual (and/or agency) and the max total fees for the full length of the contract (even if the fees will be paid periodically over the extent of the contract).
- When creating the SOW, be as specific as possible. Clearly identify the scope of the work, the acceptance criteria and the deliverables. Also consider adding criteria related to how your team works. For example, do you expect the contingent workers to add daily async updates to their issues or expect merge requests to be under x lines? If so, consider adding it to the SOW.
- Have the contingent workers or vendor representative sign the SOW.
- Procurement process must be completed. How to submit a Zip request for a contractor
- PO approved in Coupa with a confirmed engagement end date. Note: if you don't have access to Coupa, you can request access or ask somebody with access in your group to submit the request for you.
- Security and Legal Reviews must be complete in Zip.
- Background checks complete if accessing orange/red data
- If accessing orange/red data a GitLab laptop is required for your contractor. The manager must have confirmed that IT is able to deliver a laptop to the contractor's country/location.
- Look after the approval flow, optionally pinging individuals if they don't respond in a timely manner to the request.

Once the Coupa requisition is approved, the vendor/contingent workers will be able to invoice against the budget allocated at regular intervals as per the SOW. When they submit invoices, the Coupa requestor will be notified by Coupa to approve the payment.

Onboarding

With the contracting out of the way, you will want to onboard the new contingent workers into the project or into your team. This can look very different from one contingent workers to another depending on the nature of the work, the team and the project.

Generally speaking, you should know that contingent workers are referred to as Temporary Service Providers. And there is specific documentation in the handbook about their life cycle, how to create access requests for them and how to onboard and offboard them.

As a general guideline you would want them to:

- Learn about GitLab's values and culture
- Learn specifics about your team's process and rituals
- Learn about your team's technology stack
- Provision the necessary access to communication and productivity systems (project membership, ChatOps, Slack channels including production.) For this, you can rely on access requests and asking questions in ithelp

Announcements

It is important that not only your team knows about the contingent workers but there should be general awareness that Temporary Service Providers are working for GitLab.

It is encouraged to share information about new contingent workers joining in your group and stage Slack channel but you should also consider writing a short message into development, engineering-fyi Slack channels or into the Engineering Week In Review document. For these company-wide channels it is enough to mention that some contingent workers will work for a particular group.

Working with contingent workers

We treat contingent workers as regular team members when it comes to the contribution to issues or merge requests including the code review process.

SECTION: engineering/development/processes/rollout-plans.md

About This Page

This page outlines the requirements, success criteria, and how-to's of a rollout plan within the Development department.

What is a rollout plan and why do you need one?

A rollout plan is a description of how to get your change successfully applied to production and working as expected.

The process of creating a rollout plan is often more valuable than the plan itself because you will invest time in thinking about what needs to be done to achieve success. This may uncover implementation or observability problems that can be addressed before going live - when it is

often easier to make these changes.

These plans are useful to have because having a successful rollout on the first attempt reduces the possibility of rework or having your feature not operate as expected.

Creating a rollout plan

Rollout plans can be different for each project or issue that you need one for.

At a minimum, your rollout plan should:

- Provide details on the expected result.

- Provide information on how to validate that result.

- Account for any risks or barriers that may impact the rollout.

- Define a rollback plan in case the rollout is unsuccessful.

Here are some things to consider including in your rollout plan:

1. Expectations

Define what the expected outcome should be when the rollout plan is complete. How should things function? What will the user see? What details or metrics should the system provide?

Prepare for any unexpected outcomes by documenting what might happen if the rollout is unsuccessful. What signs can you look for to know? How can you reduce the risk of having these unexpected outcomes?

Document steps to take if things go wrong. This might include multiple tasks, such as engaging with the SRE team in a specific Slack channel or disabling a feature flag. This may also be a full rollback plan.

1. Metrics to observe

Provide links to data that can be monitored to understand whether your expectations are being met. Data could be based on predefined searches from tools like Sentry, Sitespeed, Grafana, Kibana, or some of our other monitoring tools.

1. Testing scenarios

Define manual tests you can take during the rollout to confirm the rollout is working as expected.

Engage with counterparts to ensure that necessary automated testing is passing.

For changes with a wide impact on the product, engage relevant product groups to gather test cases.

Account for different states of data during rollout, such as cached data or data that was in a previously valid state.

Include a pre-check list to ensure you are supporting multi-version compatibility / backwards compatibility.

1. Communication

Include a communication plan that includes relevant stakeholders - such as other stage groups, departments, or by engaging with Support to communicate changes to users.

Outline in the rollout plan locations to communicate or look for signals of an unsuccessful rollout - such as the production Slack channel or new issues list.

1. Staging and Production checks

Include any specific checks that need to be done when your changes are released on staging and production.

Include any checks specifically needed for self-managed instances. This may involve

testing the change on a reference architecture before release.

1. The rollout process itself

Describe what must happen in preparation for the rollout

Include the specific steps to follow

Include which metrics to observe and when they should be observed (for example, some features require a day of metrics to be observed before concluding if the change was successful)

1. Post rollout retro

Update any common practices for your stage/group so it's easier for the next rollout

Reflect on the rollout and share with team your learnings

Consider opening an issue/MR to automate parts of the rollout to make it safer and more efficient. Recommend to your manager that this work become part of an Engineering Allocation.

Additional rollout plan processes

There are additional rollout plan processes to keep yourself aware of:

Rolling out a low-risk feature flag

Rolling out a high-risk feature flag

Running an experiment

Rollout plan templates

These templates exist for the scenarios mentioned above, but can also be used as a base for your rollout plan:

Production change

Feature flag rollout

Experiment rollout

Examples of previous rollout plans

Preventing negative impacts on the system when there is potential to fail silently

Creating a pre-check list before rolling out a major change

Rolling out an experiment behind a feature flag

Rolling out a feature flag

SECTION: engineering/development/processes/talent-assessment.md

Talent Assessment Process

Company-wide guidelines on the Talent Assessment can be found here. The company timeline for the process remains SSOT, the guidelines below are meant to:

- Provide additional check points and due dates built into the company-wide timeline for Development specifically (as one of the larger departments at GitLab)
- Guidance on department-specific calibration section structure (attendees, leveling, etc.)
- Outline our strategy in terms of who we will be calibrating in the Development Department

Who Are We Calibrating?

Formal calibration will take place at the Senior Manager, Director, and VP levels. The thought process around who qualifies as an "outlier" for Performance and Growth Potential is outlined here.

Calibration and assessment are two different steps in the process. The assessment phase is the process of assessing each team member to determine their Performance and Growth Potential, whereas the calibration phase (occurring after initial assessments are made) is when management calibrates across the stage/org/level/etc. to discuss and align on assessments. Every team member should be assessed and have supporting points to justify those assessments - but to make calibration sessions more focused and scalable, we focus on outliers.

For Development specifically, we will calibrate:

1. Anyone in Box 1 aligned with company-wide guidelines.
1. Anyone in Box 9 who has been with GitLab longer than 3 months (Note: 3 months is measured from the kick off of the Talent Assessment cycle on 2023-10-16)
1. Anyone assessed as Exceeding for Performance
1. Anyone hired in the last 3 months not assessed as Too New To Rate
1. Anyone promoted in the Q3 promotion cycle (August 1st effective) or later not assessed as Too New To Rate
1. Anyone identified as Key Talent

It's important to note that while these will be our focus areas for calibration sessions, managers should feel free to raise any team member's assessment up for discussions if they have any questions or concerns. Calibrating outliers is not a limitation, but rather a structural adjustment to ensure this process is scalable and focused.

Note: If individual teams want to calibrate every individual, they have the ability to do this/organize/structure separately, but the due dates remain in place across the department to ensure we have enough time to review and calibrate at the various levels in the company.

How To Assess Acting/Interim

For team members who have assumed an Acting or an Interim role, we will assess team members aligned with their permanent positions (I.E. not the Acting or Interim position).

As the Talent Assessment impacts compensation, and Acting/Interim periods are not permanent, in the instance that a team member does not end up moving into the Acting/Interim role permanently, we would not want to have their compensation impacted by a temporary position.

Pre-Calibration Prep Work

In addition to the calibration session pre work on the Talent Assessment page, we ask that you complete the following:

1. Directors should conduct sub-department level calibration sessions ahead of the department-wide Development calibration sessions. In addition to calibrating on initial assessments, Directors are responsible for ensuring their respective teams assessments fall within the

expected distribution ahead of the department-wide calibration session for efficiency.

1. Performance and Growth Potential notes need to be added to the session agenda doc for each team member at least 3 business days before the synchronous calibration session.

1. Managers should include 2-3 supporting points for each team member for the assessment under each section in the agenda doc notes to help support the "why" behind the assessment. Note: The overviews in the agenda document for calibration are meant to provide enough of an overview so peers have an understanding of the "why", while simultaneously not overwhelming with information and decreasing efficiency of the session. Please limit to 2-3 supporting points per team member.

1. Key Talent assessment need to be completed and added to the session agenda doc at least 3 business days before the live calibration session.

- If an individual is indicated as Key Talent, an explanation should be added to indicate how this individual qualifies as Key Talent against our Key Talent definition. Reminder that the bar for Key Talent is set high, and that Key Talent makes up roughly ~10% of the entire population. In Development, we will be calibrating Key Talent from the Senior Manager level and above aligned with guidelines, meaning that while everyone in the organization is eligible to be identified as Key Talent, Senior Managers+ will be calibrating and making these final decisions. The rationale behind this decision is that it is important to have a holistic view of all team members when determining who meets the Key Talent criteria, which is why we require a certain scope when assessing Key Talent in the organization.

While calibrations and ultimate determinations will take place at Senior Manager+ level, initial recommendations begin in Workday and sit with team member's direct managers aligned with the process

1. Every session attendee should review the Performance/Growth Potential assessments and Key Talent overviews for outliers asynchronously ahead of the session to be prepared for live discussion/calibration.

Calibration Session Structure

Given the recent organizational changes and the creation of two new departments (Core Development and Expansion) and expansion of our Infrastructure department, calibration this year for what was the Development department will be a mix of calibration across the new departments and across what was the Development department. Given that team members and leaders spent the vast majority of the year in the Development department, we want to ensure that a level of calibration is able to take place within these groups.

To increase efficiency within the department, we will be implementing a "pre" calibration touchpoint in which small groups of Senior Managers and Directors across what was the Development department calibrate their individual contributor outliers and discuss synchronously as a group. The goal of these small group calibration sessions is to:

1. Facilitate cross-functional feedback and ensure it remains as a core part of the calibration process

1. Minimize synchronous time and maximize efficiency

1. Ensure that the approach to assessment and calibration remains consistent and calibrated across what was the Development department as we transition into new departments

The small groups are responsible for calibrating Associate, Intermediate, Senior, and Staff level individual contributor outliers across their respective teams. Engineering Manager level outliers and Principal+ individual contributors will be calibrated in larger sessions with all of the department leadership present. When determining groups we took timezone, group size, and cross-functional alignment (I.E. not in the same reporting chain) into account. Group assignments are as follows:

- Group 1: Cheryl Li and Michelle Gill
- Group 2: James Lopez, Nicole Williams
- Group 3: Crystal Poole, Dennis Tang, and Darva Satcher
- Group 4: Phil Calder and Jerome Ng
- Group 5: John Hope and Chun Du
- Group 6: Nick Nguyen, Thomas Woodham, Sam Goldstein

We recommend that groups use the Talent Assessment - Calibration Session template to ensure structure is similar to what we will use at the department level calibration sessions.

Additionally, we ask that you please discuss the following things specific to Development to ensure you are prepared for questions that may arise at the department-wide sessions:

- TO ADD

Small groups are responsible for scheduling their own calibration sessions at a convenient time for the group. The due date to complete small group calibration sessions to ensure we remain aligned to the department-wide calibration and division-wide calibration timeline is November 15, 2023 EOD PT.

Calibration Schedule

The calibration schedule below only includes calibration sessions in which department leaders and People Business Partner presence is required. Leaders are responsible for organizing the small group calibration and any sub-department calibration sessions independently ahead of the sessions below kicking off.

Session Number	Attendees	Calibration Level Focus	Session Date	Timezone Alignment	Duration	Department
-----	-----	-----	-----	-----	-----	-----
Session 1	Senior Managers+ (people managers only) + PBP	Engineering Manager level outliers	TBD	EMEA/Americas	1.5 hours	Core Development
Session 2	Senior Managers+ (people managers only) + PBP	Engineering Manager level outliers	TBD	APAC/Americas	1.5 hours	Expansion
Session 3	Senior Managers+ (people managers only) + PBP	Principal individual contributors	TBD	EMEA/Americas	1 hour	Core Development
Session 4	Director+ (people managers only) + PBP	Principal/Senior Manager level	TBD	APAC/Americas	1 hour	Expansion
Session 5	Director+ (people managers only) + PBP	Calibrate Senior Manager level	TBD	EMEA/Americas	1.5 hours	Core Development

| Session 6 | Department leaders and PBP | Calibrate Director/Distinguished level | TBD | EMEA/APAC/Americas | 2 hours | Core Development & Expansion |

Timeline

The company-wide Talent Assessment timeline can be found [here](#) and is SSOT. Below are level-specific calibration due dates embedded within the overall timeline for the Development Department.

- 2023-11-16: Sub-department level calibration and small group calibration sessions are complete
- 2023-11-20 through 2023-11-24: Development department calibration sessions with Senior Manager+ people manager attendees
- 2023-11-27: Development department calibration sessions with Director+ people manager attendees
- 2023-11-28: Development department calibration sessions with department leader + PBP

SECTION: engineering/development/processes/infra-dev-escalation/_index.md

About This Page

This page outlines the background, goals, success criteria, and implementation detail of infrastructure escalation process and Q&A.

Background

Historically there have been challenges in consistently maintaining our service level for GitLab.com customers. See the impacts in this [GitLab.com performance degradation summary document](#).

This issue is not unique to GitLab when business grows fast and the user base and workload on the hosted SaaS increases exponentially. As a result, the business growth requires corresponding changes in how we work so that customers continue experiencing best service by GitLab, which maintains and boosts our business growth momentum.

With that being said and the operation incidents recently, it becomes clear that we'll strengthen development team's DevOps practices and stand side by side with the Infrastructure team to keep GitLab.com running smoothly.

Process

To resolve GitLab.com issues faster, the development team will establish an on-call rotation and stand behind the products we deliver. During weekdays we are running a pilot in Sep/Oct of 2020 whereby an engineer in a Slack channel will be notified when they are online (in least recently escalated to order). During weekends (when most are not online) a spreadsheet will be used for escalation, with time slots required to be filled to ensure we have coverage. During the pilot program the spreadsheet will be used as a backup.

Note that the Infrastructure team keeps playing the first defense role on the frontline as usual, while they will determine if a development escalation will be initiated to get the operational issues resolved faster and more efficiently.

This new process will need development engineers on-call based on a rotation schedule. For more details, please refer to the full description of on-call process.

Goals

The on-call process was designed with the following goals in mind -

- Clear expectations, responsibility, and accountability.
- Full 24x7 coverage to shadow the Infrastructure team.
- Layered escalations to ensure SLO.
- Balanced on-duty hours of each engineer per day.
- Any individual minimally loaded throughout a calendar year.
- Flexibility of just-in-time adjustments.
- Leaving engineers in control of their own working schedules

Success Criteria

- Meet development's SLO of timely response to infrastructure escalations.
- No on-call engineer is burned out.
- Planned development work is minimally impacted.

Implementation

Refer to the Process section above for how to get started and keep running. In the spirit of iteration, the process will be continuously tuned and improved as we learn through practice.

An async retro issue will be registered and every participant is encouraged to enter feedback in the issue any time. A review will be held at 3-month checkpoint then determine next steps.

Appreciation

We want to be sure to show our appreciation for those who take support rotations, especially on the weekends. Every quarter, engineering directors and managers are encouraged to write a thank you message to all who did weekend support rotations (naming each person and how many rotations they did) in the previous quarter in the Thanks Slack channel. They are also encouraged to @ mention their managers, so they are also aware.

The reasons for doing this:

- It is not readily apparent how many weekend support rotations are taken by each engineer unless reviewed over a long time period.

- It is easy to forget to periodically do this over a long time-period unless it is part of a documented process.

- Doing this manually vs. via automation makes it more personalized and shows genuine appreciation.

Q&A

Q: Why do we need development engineers on-call?

A: In the investigation of recent performance degradation incident, it became apparent that deeper product knowledge is necessary to root cause the issue and develop sound solutions. Although infrastructure engineers are good at dealing with most incidents, it is the development engineers who are able to suggest the best short term workaround or temporary fix quickly when the issue involves deep insight into the implementation details.

Q: What efforts have been made to keep the impacts to work-life-balance minimal?

A: No engineer will be asked to work more hours than they currently work. Most of the hours they spend on-call will be days and times they would normally be working anyway. We need approximately 25% of on-call time to be used on days people wouldn't ordinarily be working, but by letting engineers choose when they do so, and not increasing total working hours, the impact of this is hopefully minimized. Engineers can also find substitutes in case of personal emergency.

Q: Can we make scheduling "smart" and dynamically page the engineer who's in regular working hours?

A: Yes. During regular weekdays we use Pagerslack to page engineers based on who is online in Slack. During the weekends/holidays we still need the spreadsheet as most do not regularly work on the weekends.

Q: Can we make it volunteer based?

A: In theory, yes. However, there are a few things to keep in mind.

What if a majority of volunteers are concentrated in nearby time zones?

If ALL volunteers are called upfront, what if this ends up with a very small group?

If volunteers are called dynamically e.g. before next rotation or month, there is ongoing administrative overhead and things can fall into cracks easily. What if no volunteer for a specific week? What if it ends up with always the same small group of people?

For weekends and holidays, we currently use a volunteer first model, and unfilled shifts are assigned. We encourage engineers to volunteer for shifts so they have more control over their schedule.

Q: What if the paged engineer doesn't carry domain expertise?

A: A layered escalation process was laid out in the process. It is also stated that first response doesn't mean solution is available right away.

An alternative was reviewed, e.g. having domain experts on-call in a similar way. This will involve more engineers and smaller on-call divisions, which will result in a more frequent shift and more on-call duties per engineer. The tradeoff was made in favor of minimizing on-call duties.

Q: Is this intended to be for both Backend Engineers and Frontend Engineers?

A: Given the fact that most of the recent issues require backend knowledge, only Backend Engineers will be involved in the first iteration. Long term, Frontend Engineers may get involved after the process is tuned and issue pattern is well understood.

Q: Why are engineers outside of the Development Department excluded from the rotation?

A: The rotation is intended to provide access to engineers that are directly involved with product development to stand behind the products that we deliver. Engineers in other departments have different responsibilities. For example, those in Infrastructure must focus on the scaling and reliability aspects of GitLab.com, since part of their job description is to work towards achieving GitLab.com SLOs. And those in Engineering Productivity are focused on providing tooling, and this is somewhat removed from product development.

Q: How do we answer interview candidates when they ask about on-call?

A: Let's describe the full picture of our incident handling model and tell candidates there are chances development engineers will be on-call and assist resolving GitLab.com operational incidents.

Usually, the infrastructure team plays the first defense role on the frontline. Development engineers will only be called when the infrastructure team determines that development escalation is necessary.

Q: Will our job description be updated?

A: The engineering leadership team will work with the talent acquisition team on this. Should there be any changes to job descriptions, all interviewers will be informed.

Q: What if my local law restricts or prohibits on-call duty?

A: All are encouraged to share their local law and regulation information with their manager and will not be scheduled until further clarification is obtained should there be any concern or ambiguity of the local law and regulation. Engineering leadership team will also work with Legal and PeopleOps to obtain clarification with regard to this. We maintain a list of rules in a Google Doc.

Q: Did we consider using PagerDuty?

A: Yes, we did. It was decided to keep it lightweight with Slack in the first round experiment, because there is work to enhance the chatops bot.

Q: What are the expectations for my existing work while I'm on-call?

A: While on-call the expectation of existing work is that it is effectively suspended. Managers are required to plan for on-call engineers to be unavailable. If you are able to make progress because there are no ongoing incidents that is welcomed, but work must stop if an on-call request is made.

Q: What is the expectation for escalations which are still open at the boundaries of an on-call shift?

A: Similar to bullet 7 under Guidelines section - Relay Handover, summarize the status and investigations by far, then handover.

Q: Can Slack only be configured to trigger notification from infra-escalation out-of-hours, especially during hours of 0400-0700 (APAC) ? I already receive lots of pings out-of-hours but this don't wake me up currently as I have Do-not-disturb turned on.

A: It seems notifications can be customized on mobile app, check out this guide (Android device)

Mute all other channels but the escalation channel during a specific time period (i.e. 4-8am) through the Channel-specific notifications setting on Notifications screen, which may help reduce noise and only get pinged for escalations.

Q: Is there any concept of compensation? This can be in any form (pay, time off, etc)

A: On-call work can be considered a deliverable like any other. It doesn't imply working any extra hours - but a few hours will be at less desirable times than now. Although no compensation changes are anticipated to account for this, we may consider discretionary rewards for people who exceed expectations when choosing less-desirable hours.

Q: How will the volume of escalations to the on-call engineer be measured? Have we established thresholds to know when a working group may need to be established to remediate a "hot" set of issues?

A: Let's start with hand counting and review the volume at the Infra/Dev meeting. This can also be added to the board.

Q: We are discussing the concept of working hours for new-on-call and having expected shifts, however, this is a departure from non-on-call based on this in the handbook. /handbook/values/measure-impact-not-activity Is this an intentional policy shift?

A: This is not a shift in policy. Engineers are still in control of their schedules, and can choose when to work, as long as the overall goal of full coverage of the rotation is met. The policy of impact vs. activity is based on delivering functionality. On call is about addressing operational issues which can happen at any time and need to be addressed immediately. So the policies are congruent.

Q: In order to effectively debug production issues, developers may require expanded access to production systems and metrics. Is the plan for developers to be on-call solely for consultative purposes without need for direct debugging of systems? If they need access to production systems how will they be onboarded?

A: For the first iteration, the plan is consultative and if any code changes are required the oncall makes them. Direct debugging is not required and it is expected infrastructure can relate production issues effectively to the oncall for progress to be made. We are not planning onboarding to production at this time.

Q: If we had this process in place for recent outages, would it have resolved them significantly faster? i.e. is the gap to find an engineer to support currently our biggest problem?

A: If you look at the chart of outages as part of the Performance degradation (see above link), you will see outages on the June 5th, July 1st, 3rd. Had we caught the issues and worked to address them on June 5th we could have prevented July 1st degradation. July 3rd is half degradation, half attack. So we would have minimized some level of impact here as well. The time associated with this degradations is also high (540 minutes for July 1st) and we could have reduced that time as well.

Q: Isn't this more about discipline of seeing incidents through to resolution, not just how quickly we respond to them? It feels the on-call process addresses the latter, but not former.

A: It's actually both and we are working to address both. We have also added a Infra/Dev issue board to track concerns to see through to resolution and make sure we have the right priority and severity on them. It's likely that oncall escalations will end up with follow on items for this board. This page gives the description.

Q: How should the infrastructure member make international phone calls to page engineers?

A: Zoom supports making international calls with low rates. This can be done from inside an ongoing Zoom call under Invite > Phone. Considering that this will only be used for a quick phone call, to alert the engineer of an ongoing escalation, the cost for GitLab will be very minimal.

SECTION: engineering/development/processes/infra-dev-escalation/process/index.md

{{% alert title="Quick escalation guide" color="danger" %}}

Please use /devoncall <incidenturl> on Slack for any escalation that meets the criteria.

{{% /alert %}}

{{% alert title="Note" color="primary" %}}

We use Pagerslack for weekday escalations. Weekend escalations rely on the spreadsheet solely.

{{% /alert %}}

About This Page

This page outlines the development team on-call process and guidelines for developing the rotation schedule for handling incident escalations.

Expectation

The expectation for the development engineers is to be a technical consultant and collaborate with the teams who requested on-call escalation to troubleshoot together. There is no expectation that the development engineer is solely responsible for a resolution of the escalation.

Escalation Process

Scope of Process

1. This process is designed for the following issues:
 1. GitLab.com and self-managed hosting operational emergencies raised by the Infrastructure, Security, and Support teams.
 1. Engineering emergencies raised by teams in engineering such as the Delivery and QE teams, where an imminent deployment or release is blocked.
 1. Engineering emergencies raised by teams in product such as the Product Operations team, where the imminent deployment of the Release Post is blocked.
 1. Engineering emergencies raised by teams in quality such as the Engineering Productivity team, where merge requests merges are blocked due to the master branch failing.
1. This process is NOT a path to reach development team for non-urgent issues that the Infrastructure, Security, and Support teams run into. Such issues can be moved forward by:
 1. Labelling with security and the @gitlab-com/security/appsec team mentioned to be notified as part of the Application Security Triage rotation
 1. Labelling with infradev which will be raised to Infra/Dev triage board
 1. Raising to the respective product stage/group Slack channel, or
 1. Asking the is-this-known Slack channel
1. This process provides 24x7 coverage.

Example of qualified issue

1. Production issue examples:
 1. GitLab.com: DB failover and degraded GitLab.com performance
 1. GitLab.com: Severity 1 vulnerability being actively exploited or high likelihood of being exploited and puts the confidentiality, availability, and/or integrity of customer data in jeopardy.
 1. Self-managed: Customer ticket - Our GitLab instance is down - request an emergency support
 1. Self-managed: Customer ticket - Database overloaded due to Sidekiq stuck jobs
1. Engineering emergency examples:
 1. A post deployment issue with version.gitlab.com that will cause self-managed deployment failure.
 1. GitLab.com deployment or a security release is blocked due to pipeline failure.
 1. A severity::1 regression found in the release staging.gitlab.com
 1. Evaluating severity and making decision on a potential release or deployment show stopper collaboratively when other engineering teams are challenged to accomplish alone.
 1. master pipelines are broken.

Examples of non-qualified issues

1. Production issue examples:
 1. GitLab.Com:Errors when import from GitHub
 1. GitLab.com: Last minute security patch to be included in an upcoming release
 1. Self-managed(ZD): View switch causing browser freeze
 1. Self-managed(ZD): Watch Everything Notification Level
1. Engineering issue examples:
 1. A priority::1/severity::1 enhancement of CI

1. A priority::1/severity::1 fix to API
1. Non release blocking QA failures on staging.gitlab.com

Process Outline

NOTE: On-call engineer do not need to announce the beginning and end of their shift in dev-escalation unless there is an active incident happening (check the chat history of the channel to know if there is an active incident). This is because many engineers have very noisy notifications enabled for that channel, and such announcements are essentially false positives which make them check the channel unnecessarily.

Weekdays (UTC)

Weekdays will now leverage an automated system relying upon the Pagerslack.

1. Incidents will be escalated by Pagerslack and randomly select a Backend Engineer (BE) that is currently online in Slack and eligible due to their working hours.
1. Pagerslack would also factor recency into the algorithm so that BEs that had recently been involved in an incident would be selected last.
1. During incidents a randomly selected BE has the option to pass the incident to another BE if they are urgently needed somewhere else.
1. Engineers who are eligible to be on-call during weekend shifts are deprioritized from the process

Escalation

1. SRE et al, types /devoncall incident-issue-url into dev-escalation
1. Pagerslack randomly selects a Backend Engineer (BE) first responder based on: working hours, whether they are online and notifies them via slack/cell etc.
1. BE responds to the Pagerslack thread with ·
1. If Primary does not respond a secondary will be notified.
1. Pagerslack will continue trying up to 6 different BEs with a preference for those who do not take weekend shifts
1. BE triages the issue and works towards a solution.
1. If necessary, BE reach out to domain experts as needed.

In the event that no BE engineers respond to the bot, Pagerslack will then notify the Engineering Managers. They will need to find someone available and notify this in the escalation thread. As an EM:

1. Check whether some of the engineers pinged belong to your group and see whether they are available to help
1. Try to find someone available from your group
1. If the search is positive, leave a message in the thread as an acknowledgement that the engineer from your group will be looking into the issue

Weekends and Holidays (UTC)

Weekends/Holidays are included in the On-call spreadsheet. Those holidays include: Family & Friends days, and any major holidays that affect multiple timezones.

There are restrictions of eligibility in certain localities, which can be found in the weekend column of the Development Team BE spreadsheet spreadsheet. This spreadsheet is refreshed by the end of each month automatically via the Employment Automation.

Escalation

1. SRE et al, types /devoncall incident-issue-url into dev-escalation
1. Pagerslack posts a link to the On-call spreadsheet
1. SRE contacts the scheduled BE via Slack or make a phone call 5 minutes after Slack ping. Refer to bullet 3 under process-outline.
1. BE triages the issue and works towards a solution.
1. If necessary, BE will reach out to domain experts as needed.

Reaching out to domain experts

NOTE: these people generally do not carry pagers and are only likely available during business times. Please respect the individual intake processes, this helps each team track their work.

Use the product sections handbook page to determine whom to contact.

- Get attention from Gitaly team: issues related to Git and Git repositories, Gitaly Cluster, including data recovery.
- Get attention from Fulfillment team: issues related to CustomersDot (customers.gitlab.com).

First response time SLOs

OPERATIONAL EMERGENCY ISSUES ONLY

1. GitLab.com: Development engineers provide initial response (not solution) in both dev-escalation and the tracking issue within 15 minutes.
1. Self-managed: Development engineers provide initial response (not solution) in both dev-escalation and the tracking issue on a best-effort basis. (SLO will be determined at a later time.)
1. In the case of a tie between GitLab.com and self-managed issues, GitLab.com issue takes priority.
1. In the case of a tie between production (GitLab.com, self-managed) and engineering issues, production issue takes priority. The preferred action is to either backout or rollback to the point before the offending MR.

Required Slack Channel and Notification Settings

1. All on-call engineers, managers, distinguished engineers, fellows (who are not co-founders) and directors are required to join dev-escalation.
1. On-call engineers are required to add a phone number that they can be reached on during their on-call schedule to the On-call spreadsheet.
1. On-call engineers are required to turn on Slack notification during regular working hours. Please refer to Notification Settings for details.
1. Similarly, managers and directors of on duty engineers are also recommended to do the same above to be informed. When necessary, managers and directors will assist to find domain experts.

1. Hint: turn on Slack email notification while on duty to double ensure things don't fall into cracks.

Rotation Scheduling

Important: Sign-ups of weekends/holidays are required as a backup while using Pagerslack as the primary escalation tool.

Guidelines

Assignments

On-call work comes in four-hour blocks, aligned to UTC:

- 0000 - 0359
- 0400 - 0759
- 0800 - 1159
- 1200 - 1559
- 1600 - 1959
- 2000 - 2359

One engineer must be on-call at all times on weekends and company holidays. This means that each year, we must allocate 756 4-hour shifts.

The total number of shifts is divided among the eligible engineers. This is the minimum number of shifts any one engineer is expected to do. As of February 2024 we have around 206 eligible engineers, this means each engineer is expected to do 4 shifts per year, or 1 shift per quarter.

In general, engineers are free to choose which shifts they take across the year. They are free to choose shifts that are convenient for them, and to arrange shifts in blocks if they prefer. A few conditions apply:

- No engineer should be on call for more than 3 shifts in a row (12 hours), with 1-2 being the norm
- No engineer should take more than 12 shifts (48 hours) per week, with 10 shifts (40 hours) being the usual maximum.

Scheduling and claiming specific shifts is done on the On-call spreadsheet.

Scheduling spreadsheets

The following internal-only scheduling spreadsheets are maintained as part of the development team on-call process:

- [On-call spreadsheet](#)
- [Development Team BE spreadsheet](#)
- [IMOC Rotation emails spreadsheet](#)
- [Excluded Team Member Emails](#)

- [Summit Availability](#)

Eligibility

All backend and fullstack engineers within Core Development, Expansion Development, or Core Platform who have been with the company for at least 3 months, with exceptions recorded in the Excluded Team Member Emails spreadsheet.

Other general exceptions: (i.e. exempted from on-call duty)

- Distinguished engineers and above.
- Intern and Associate engineers.
- Where the law or regulation of the country/region poses restrictions. According to legal department -
 - There are countries with laws governing hours that can be worked.
 - This would not be an issue in the U.S.
 - At this point we would only be looking into countries where 1) we have legal entities, as those team members are employees or 2) countries where team members are hired as employees through one of our PEO providers. For everyone else, team members are contracted as independent contractors so general employment law would not apply.
- Team members participating in another on-call rotation. Examples of this include a team specific rotation for a service owned by that team or in the Engineering Incident Manager rotation.

The eligibility is maintained in this Development Team BE spreadsheet and part of the spreadsheet is refreshed automatically by our automation script.

Nomination

Engineers should claim shifts themselves on the On-call spreadsheet.

To ensure we get 100% coverage, the schedule is fixed one month in advance.

Engineers claim shifts between two and three months in advance. When signing up, fill the cell with your full name as it appears in the team members list (internal only), GitLab display name, and phone number with country code. This same instruction is posted on the header of schedule spreadsheet too.

At the middle of each month, engineering managers look at the schedule for the following month (e.g. on the 15th March, they would be considering the schedule for April, and engineers are claiming slots in May). If any gaps or uncovered shifts are identified, the EMs will assign those shifts to engineers. The assignment should take into account:

- How many on-call hours an engineer has done (i.e., how many of their allocated hours are left)
- Upcoming leave
- Any other extenuating factors
- Respecting an assumed 40-hour working week
- Respecting an assumed 8-hour working day
- Respecting the timezones engineers are based in
- Ensuring assigned on-call shifts fit between the hours of 8am-8pm local time for all team

members in general, and 7am-9pm for engineers in the US Pacific Time timezone (approved in this issue).

- Optimize for better alignment of shift hours

- When assigning the first Friday shifts (00:00 UTC/Saturday 1pm NZT/ 4-6pm PST). Please favour/allocate these shifts to AMER engineers within their working hours.

- When assigning the final Sunday shift of a typical weekend 20:00-00:00 UTC/Sunday Monday NZT, choose a New Zealand based engineer unless it is a public holiday in New Zealand.

In general, engineers who aren't signing up to cover on-call shifts will be the ones who end up being assigned shifts that nobody else wants to cover, so it's best to sign up for shifts early!

There is additional information regarding weekend shifts, which can be found in "Additional Notes for Weekend Shifts"(internal only) under a sub-folder called Development Escalation Process in the shared Engineering folder on Google Drive.

Relay Handover

- Since the engineers who are on call may change frequently, responsibility for being available rests with them. Missing an on-call shift is a serious matter.

- In the instance of an ongoing escalation, no engineer should finish their on-call duties until they have paged and confirmed the engineer taking over from them is present, or they have notified someone who is able to arrange a replacement. They do not have to find a replacement themselves, but they need confirmation from someone that a replacement will be found.

If the shift for the current on-call dev is ending, and the incoming on-call dev listed in the On-call spreadsheet is not responding, please use the following process to find an engineer to take over the shift, stopping the process as soon as an engineer is found:

1. Call the phone number for the incoming on-call dev listed in the On-call spreadsheet to see if they're able to fulfill their assigned shift.

1. Get in touch with the next on-call dev listed in the On-call spreadsheet, if it's within their working hours, to see if they're able to take on both their assigned shift, as well as the shift from the unavailable incoming on-call dev.

1. Post a message to the existing incident thread in the dev-escalation channel, or create a new thread requesting volunteers to take over the shift.

1. Message the on-call IMOC and ask them for help finding an engineer to take over the shift.

- In the instance of an ongoing escalation being handed over to another incoming on-call engineer the current on-call engineers summarize full context of on-going issues, such as but not limited to

- Current status

- What was attempted

- What to explore next if any clue

- Anything that helps bring the next on-call engineer up to speed quickly

These summary items should be in written format in the following locations:

- Existing threads in dev-escalation
- Incident tracking issues

This shall be completed at the end of shifts to hand over smoothly.

- For current Infrastructure issues and status, refer to Infra/Dev Triage board.
- For current Production incident issues and status, refer to Production Incidents board.
- If an incident is ongoing at the time of handover, outgoing engineers may prefer to remain on-call for another shift. This is acceptable as long as the incoming engineer agrees, and the outgoing engineer is on their first or second shift.
- If you were involved in an incident which has been mitigated during your shift, leave a note about your involvement in the incident issue and link to it in the dev-escalation Slack channel indicating you participated in the issue as an informational hand-off to future on-call engineers.

Development on-call DRI

Current DRI: James Lopez

The development on-call DRI has the following responsibilities:

1. Ensure the process and this handbook page is up to date and promptly fix any problems or answer any queries
1. Select coordinators for the monthly schedule (at least 15 days before the new month starts)
 - a tool can be used to assist with this
1. Keep the IMOC Rotation emails spreadsheet up to date and export them monthly from Pagerduty.
1. Keep the Excluded Team Member Emails spreadsheet up to date.
1. Keep the On-call spreadsheet up to date, including Family and Friends days, other general public holidays, Total Shift counts, and escalation tracking
1. Keep the Summit Availability spreadsheet up to date.
1. Monitor dev-escalation and dev-oncall on Slack and resolve any problems with the escalation bot and the PeopleOps integration
1. Track, document, and keep up to date eligibility criteria as well as onboarding/offboarding and remind Engineering managers about this process
1. Implement an appreciation process for engineers who were on-call and handled recent incidents

Coordinator

Given the complexity of administration overhead, one engineering director or manager will be responsible to coordinate the scheduling of one month. The nomination follows the same approach where self-nomination is preferred. If there are no volunteers, the DRI will select them for upcoming months.

The coordinators should assign themselves to the corresponding monthly issue from this epic.

Responsibility

The coordinator will:

1. Remind engineers to sign up, by:
 - Posting reminders to the development and backend channels in Slack
 - Asking managers in eng-managers to remind team-members in 1-1s
1. Assign folks to unfilled slots when needed. Use purple text in the spreadsheet to indicate this was an assigned slot.
1. Ensure that all developers listed in the On-call spreadsheet have:
 - Provided a phone number
 - Confirmed their shift by changing the font color of their entry from purple to black.
1. Coordinate temporary changes or special requests that cannot be resolved by engineers themselves. Fix any schedules that conflict with local labor law.
1. After assigning unfilled slots and accommodating special requests the coordinator should click Sync to Calendar > Schedule shifts.
This will schedule shifts on the On-call schedule shifts calendar and if any developer added their email into the spreadsheet, they will be added as guests in the on-call calendar event. Ensure that you have subscribed to the calendar before syncing.

An Epic of execution tracking was created, where each coordinator is expected to register an issue under this Epic for the month-on-duty to capture activities and send notifications. Here is an example.

How-To

Refer to this coordinator issue template for instructions or the steps below. The same template is used for monthly on-call planning and execution issues.

An automated process can be followed. If there are problems with the tool, a manual process can be followed:

1. Start by finding the least filled shift (Usually this is 00:00 - 04:00 UTC) in the On-call spreadsheet.
1. Determine the appropriate timezone for this shift (in the case of 00:00 - 04:00 it is +9,+10,+11,+12,+13).
1. Go to the Development Team BE spreadsheet and filter the "timezone" column by the desired timezones for the shift. Now you have the list of possible people that can take this shift. Alternatively, you can use this dev-on-call tool to find out people who may take this shift.
1. Go to google calendar and start to create a dummy event that is on the day and time of the unclaimed shift . NOTE you will not actually end up creating this event.
1. Add all of the people that can possibly take the shift to the event as guests.
1. Go to the "Find a Time" tab in the calendar event to see availabilities of people.
1. Find a person that is available (preferring people that have taken less than 4 shifts per quarter and few or no shifts overall, based on the shifts counts sheet (internal only)). Note people who are on leave or otherwise busy or in interviews, do not schedule them for the shift. It would be fine to ignore events that appeared to be normal team meetings, 1:1, coffee chat as people can always leave a meeting if there is an urgent escalation.
1. Assign them to the shift by filling their name in the On-call spreadsheet in Purple font color.

1. Now since there are likely many days that have this unfilled time slot then update the event date to the next day with this same unfilled time zone. Since it's the same time then the same set of people will be appropriate to take the shift which means you don't need to update the guest list.

1. Repeat all of the above for all of the unclaimed timezones remembering that you want to solve for one shift (by time range) at a time as it means you will re-use the same guest list to determine availability.

Additional Notes for Weekend Shifts

For those eligible engineers, everyone is encouraged to explore options that work best for their personal situations in lieu of weekend shifts. When on-call you have the following possibilities:

1. Swap weekend days and weekdays.

1. Swap hours between weekend days and weekdays.

1. Take up to double the time off for any time worked during the weekend when the above two options don't work with your personal schedule.

1. When an engineer is in standby mode (e.g. not paged) during the weekend shift, they can take 1.25x time-off.

1. When an engineer is in call-back mode (e.g. being paged) during the weekend shift, they can take double the time-off.

1. For those who reside in Australia, please refer to these guidelines of time in lieu in the handbook.

1. Please create an OOO event in Workday and choose On-Call Time in Lieu.

1. Other alternatives that promote work-life balance and have the least impact to your personal schedule.

It is important to abide by local labor laws and you are encouraged to understand if there are restrictions around your working time. The purpose of this information is to encourage you to take time off according to your schedule to account for the interruption to your weekend.

If you prefer to work on a preferred weekend day please proactively sign up for shifts to avoid auto-assignment. Team members who have signed up for the fewest shifts are auto-assigned open shifts first.

Rotation Schedule

See the On-call spreadsheet.

Resources

Responding Guidelines

Infrastructure or availability incident

When responding to an Infrastructure Incident, utilize the below procedure as guidelines to follow to assist both yourself and the members requesting your assistance

1. Respond to the slack page with the :eyes: reaction - this signals to the bot that you are

looking into the problem

1. Join the appropriate incident slack channel for all communications that are text based - Normally this is inc-<NUMBER>

1. Join the Incident Zoom - this can be found bookmarked in the relevant incident Slack Channel

1. Work with the EOC to determine if a known code path is problematic

- Should the knowledge of this be in your domain, continue working with the Engineer to troubleshoot the problem

- Should this be something you may be unfamiliar with, attempt to determine code ownership by team - Knowing this will enable us to see if we can bring online an Engineer from that team into the Incident

1. Work with the Incident Manager to ensure that the Incident issue is assigned to the appropriate Engineering Manager - if applicable

Master broken guidelines

You can read about the process workflow here and follow the triage responsibilities

Bot (Pagerslack) Usage

- In order to use the bot, type /devon-call incident-issue-url to trigger the escalation process.

- Message the bot privately with top to show the top 25 members that are next in the escalation queue

- Message the bot privately with position to see your position in the queue. The higher the number, the less probabilities to get pinged.

Please report any problems by creating an issue in the pagerslack project.

Shadowing An Incident Triage Session

Feel free to participate in any incident triaging call if you would like to have a few rehearsals of how it usually works. Simply watch out for active incidents in incidents-dotcom and join the Situation Room Zoom call (link can be found in the channel) for synchronous troubleshooting. There is a nice blog post about the shadowing experience.

Replaying Previous Incidents

Situation Room recordings from previous incidents are available in internal Google Drive folder.

Shadowing A Whole Shift

To get an idea of what's expected of an on-call engineer and how often incidents occur it can be helpful to shadow another shift. To do this simply identify a time-slot that you'd like to shadow in the On-call spreadsheet and contact the primary to let them know you'll be shadowing. Ask them to invite you to the calendar event for this slot. During the shift keep an eye on dev-escalation for incidents and observe how the primary follows the process if any arise.

Tips & Tricks of Troubleshooting

1. How to Investigate a 500 error using Sentry and Kibana.
1. Walkthrough of GitLab.com's SLO Framework.
1. Scalability documentation.
1. Use Grafana and Kibana to look at PostgreSQL data to find the root cause.
 - Related incident: Postgres transactions timing out; sidekiq queues below apdex score; and overdue pull mirror jobs.
1. Use Grafana and Prometheus to troubleshoot API slowdown.
 - Related incident: 2019-11-27 Increased latency on API fleet.
1. Let's make 500s more fun

Tools for Engineers

1. Training videos of available tools
 1. Visualization Tools Playlist.
 1. Monitoring Tools Playlist.
 1. How to create Kibana visualizations for checking performance.
1. Dashboards examples, more are available via the dropdown at upper-left corner of any dashboard below
 1. Saturation Component Alert.
 1. Service Platform Metrics.
 1. SLAs.
 1. Web Overview.

Help with the process

If you have questions about the process, please reach out to dev-oncall in Slack or to the DRI directly.

Pagerslack statistics

Tableau

```
{{< tableau height="600px" src="https://us-west-2b.online.tableau.com/t/gitlabpublic/views/PagerslackStats/ofEscalationsperMonth" >}}  
{{< /tableau >}}
```

```
{{< tableau height="600px" src="https://us-west-2b.online.tableau.com/t/gitlabpublic/views/PagerslackStats/Responsetime" >}}  
{{< /tableau >}}
```

```
{{< tableau height="600px" src="https://us-west-2b.online.tableau.com/t/gitlabpublic/views/PagerslackStats/Percentageofbotescalatedresponses" >}}  
{{< /tableau >}}
```

Notification Settings

To make the First Responder process effective, the engineer on-call must configure their

notifications to give them the best chance of noticing and responding to an incident.

These are the recommended settings. Your mileage may vary.

Slack Notifications

1. Within Slack, open "Preferences".
 1. Under "Notify me about...", select one of the first two options; we recommend "Direct messages, mentions & keywords". Do not choose "Nothing".
 1. If you check "Use different settings for my mobile devices", follow the same rule above.
 1. Scroll down to "Notification Schedule".
 1. Under "Allow notifications", enter your work schedule. For example: Weekdays, 9 am to 5 pm. Pagerslack relies on this to decide whether or not to page a person.
 1. Scroll down to "Sound & appearance".
 1. Choose settings that ensure you won't miss messages. We recommend:
 1. Select a "Notification sound".
 1. Check "Bounce Slack's icon when receiving a notification".
 1. Use your preference for the other settings. The "Channel-specific notifications" are particularly helpful to mute noisy channels that you don't need to be interrupted for.

macOS Notifications

1. Under "System Preferences", select "Notifications".
 1. Scroll down to find "Slack".
 1. Enable "Allow Notifications from Slack".
 1. For "Slack alert style", we recommend "Alerts" so you need to dismiss them. "Banners" might also work for you. Do not select "None".
 1. Enable "Play sound for notifications", particularly if you chose "Banners" above.
 1. Use your preference for the other settings.

iOS Notifications

1. Under "Settings", open "Notifications".
 1. Scroll down to find "Slack".
 1. Enable "Allow Notifications from Slack".
 1. Under "ALERTS", enable "Lock Screen" and "Banners".
 1. Enable "Sounds".
 1. Use your preference for the other settings.

SECTION: engineering/development/sec/_index.md

<div class="diagramwrapper">

<div class="box"><svg xmlns="https://www.w3.org/2000/svg" viewBox="0 0 1025 500" role="img"

aria-label="GitLab DevSecOps Lifecycle">

<style>

```
.text {
  font-family: Inter,Source Sans Pro,sans-serif;
  font-size: 20px;
  font-weight: bold;
  fill: 2b2838;
  letter-spacing: .02em;
  dominant-baseline: middle;
  text-anchor: middle;
}
.text2 {
  font-size: 26px;
  fill: 2b2838;
}
.devsecops {
  font-size: 56px;
  fill: black;
}
.mix {
  mix-blend-mode: color-burn;
}
.line {
  fill: none;
  stroke: 2b2838;
  stroke-width: 2px;
}
.highlighted {
  font-size: 22px;
}
.highlightedline {
  stroke-width: 3px;
}
.whitefill {
  fill: fff;
}
.faded {
  fill: a09fa4;
}
.fadestroke {
  stroke: a09fa4;
}
.lightline{
  stroke: d3d0d4;
  stroke-width: 3px;
}
```

</style>
<linearGradient id="linear-gradient" x1="18" y1="250" x2="1007" y2="250"
gradientTransform="matrix(1, 0, 0, 1, 0, 0)" gradientUnits="userSpaceOnUse">
<stop offset="0" stop-color="e6dcf4"></stop>
<stop offset="1" stop-color="ffd0be"></stop>

```

</linearGradient>
<linearGradient id="linear-gradient-2" x1="720.88" y1="337.91" x2="370.24" y2="74.18"
gradientTransform="matrix(1, 0, 0, 1, 0, 0)" gradientUnits="userSpaceOnUse">
<stop offset="0" stop-color="2b2838" stop-opacity=".24"></stop>
<stop offset="1" stop-color="45424d" stop-opacity="0"></stop>
</linearGradient>
<linearGradient id="linear-gradient-3" x1="304.12" y1="162.09" x2="654.76" y2="425.82"
gradientTransform="matrix(1, 0, 0, 1, 0, 0)" gradientUnits="userSpaceOnUse">
<stop offset="0" stop-color="2b2838" stop-opacity=".24"></stop>
<stop offset="1" stop-color="45424d" stop-opacity="0"></stop>
</linearGradient>
<path id="background" fill="fff" d="M 0,0 L 1025,0 L 1025,500 L 0,500"></path><path
id="upperarc" class="line highlightedline" d="M 0,250 a238,238 0 0,1 238,-238 L 787,12 a238,238
0 0,1 238,238"></path><path id="lowerarc" class="line highlightedline" d="M 0,250 a238,238 0
0,0 238,238 L 787,488 a238,238 0 0,0 238,-238"></path><path id="clear1" fill="fff" d="M -3,190
L -3,310 L 1028,310 L 1028,190"></path>
<path id="clear2" fill="fff" d="M 422,0 L 603,0 L 603,500 L 422,500"></path>
<path id="loop" fill="url(linear-gradient)" d="M 512.5,181.18 L 370.24,74.18 a220,220 0 1,0
0,351.64 L 512.5,318.8 L 654.76,425.82 a220,220 0 1,0 0,-351.64"></path>
<path id="cutout1" fill="fff" d="M 421,250 L 304.12,162.09 a110,110 0 1,0 0,175.82"></path>
<path id="cutout2" fill="fff" d="M 604,250 L 720.88,162.09 a110,110 0 1,1 0,175.82"></path>
<path id="shadow1" fill="url(linear-gradient-2)" class="mix" d="M 512.5,181.18 L 421,250 L
304.12,162.09 L 370.24,74.18"></path>
<path id="shadow2" fill="url(linear-gradient-3)" class="mix" d="M 512.5,318.8 L 604,250 L
720.88,337.91 L 654.76,425.82"></path><path class="lightline" fill="none" d="M 421,256 L
304.12,168.09 a104,104 0 1,0 -64.12,185.91 L 225,341"></path>
<path class="lightline" fill="none" d="M 604,256 L 720.88,168.09 a104,104 0 1,1 64.12,185.91 L
800,341"></path>
<path fill="fff" class="lightline" d="M 357,205 L 668,205 a45,45 0 0,1 0,90 L 357,295 a45,45 0
0,1 0,-90"></path>
<path d="M536.514 419L536.448 418.82L529.663 401.124C529.526 400.778 529.281 400.483 528.965
400.283C528.73 400.131 528.461 400.039 528.184 400.01C527.905 399.981 527.624 400.0

```